

A(nother) geometric comparison between representations of real and p-adic groups

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Postdoc Workshop IV

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- G - connected reductive algebraic group $/\mathbb{C}$
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Infinitesimal character:

- $\mathfrak{g}$ - Lie algebra of G
- $\text{Rep}(G_{\mathbb{R}}) \rightarrow \text{Mod}(\mathfrak{g}) \cong \text{Mod}(\mathcal{U}(\mathfrak{g})) \twoheadrightarrow \text{Mod}(Z(\mathcal{U}(\mathfrak{g})))$
- $\Lambda_{\mathbb{R}} : Z(\mathcal{U}(\mathfrak{g})) \rightarrow \mathbb{C}$ - *infinitesimal character*

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$\Pi(G_{\mathbb{R}}, \Lambda_{\mathbb{R}})$ - irreducible representations of $G_{\mathbb{R}}$ with infinitesimal character $\Lambda_{\mathbb{R}} / \sim$

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$$\Pi(G_{\mathbb{R}}, \Lambda_{\mathbb{R}}) \approx \{(\text{enhanced}) \text{ Langlands parameters}\} / \sim .$$

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Adams-Barbasch-Vogan:

$$\{(\text{enhanced}) \text{ Langlands parameters}\} / \sim \approx \text{Irr Perv}(\check{B}, \check{K})$$

(under some conditions...)

Langlands parameter space $/\mathbb{R}$

- $\check{G}$ - Langlands dual group
- $\check{B} \subset \check{G}$ - Borel subgroup (upper-triangular matrices)
 $\rightsquigarrow \check{B} = \check{G}/\check{B}$ - flag variety of $\check{G}$
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$\text{Perv}(\check{B}, \check{K})$ - category of $\check{K}$ -equivariant perverse sheaves on $\check{B}$

$\rightsquigarrow \text{Irr Perv}(\check{B}, \check{K})$ - irreducible objects

SL_2 example

$$G_{\mathbb{R}} = \mathbf{PGL}_2(\mathbb{R}), \quad G = \mathbf{PGL}_2(\mathbb{C}),$$

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$$\rightsquigarrow \Pi(\mathbf{PGL}_2(\mathbb{R}), trv) = \{$$

$$\mathbb{C}^+ \text{ (1-dim'l)}$$

$$\mathbb{C}^- \text{ (1-dim'l)}$$

$$DS \text{ } (\infty\text{-dim'l})$$

$$\}$$

$\mathbf{SL}_2$ example

$$\check{G}_{\mathbb{R}} = \mathbf{SL}_2(\mathbb{R}), \quad \check{G} = \mathbf{SL}_2(\mathbb{C}),$$

$$\check{K} = \mathbb{C}^{\times} \hookrightarrow \mathbb{P}^1 = \check{B}$$

$$\rightsquigarrow \mathrm{Irr} \, \mathrm{Perv}(\mathbb{P}^1, \mathbb{C}^{\times}) = \{$$

skyscraper at $0 \in \mathbb{P}^1$

skyscraper at $\infty \in \mathbb{P}^1$

constant sheaf $\underline{\mathbb{C}}_{\mathbb{P}^1}$

$\mathcal{I}$ = “Möbius strip over $\mathbb{C}^* \subset \mathbb{P}^1$ ”

$$\}$$

SL_2 example

$\Pi(\mathbf{PGL}_2(\mathbb{R}), trv)$	$\text{Irr Perv}(\mathbb{P}^1, \mathbb{C}^\times)$
$\mathbb{C}^+$	skyscraper at 0
$\mathbb{C}^-$	skyscraper at ∞
DS	$\underline{\mathbb{C}}_{\mathbb{P}^1}$
$(\mathbb{C} \text{ of } \mathbf{PSU}(2))$	$\mathcal{I}$

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Vogan:

$$\{(\text{enhanced}) \text{ Langlands parameters}\} / \sim \cong \text{Irr Perv}(X_{\Lambda_p}, Z_{G'_p}(\Lambda_p))$$

Langlands parameter space $/\mathbb{Q}_p$

$$X_{\Lambda_p} = \{ \xi \in \mathfrak{g}'_p \mid \forall w \in W_{\mathbb{Q}_p}, \text{Ad}(\Lambda_p(w))\xi = |w|\xi \}$$

(conical affine variety in $\mathfrak{g}'_p$)

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Goal

Find pairs of groups $(G_{\mathbb{R}}, G'_p)$ and pairs of infinitesimal characters $(\Lambda_{\mathbb{R}}, \Lambda_p)$ so that

$$\Pi(G'_p, \Lambda_p) \sim \Pi(G_{\mathbb{R}}, \Lambda_{\mathbb{R}})$$

Idea

Instead of trying to directly build a map

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$$X_{\Lambda_p} \leftarrow \check{\mathcal{B}}$$

In practice:

$$[X_{\Lambda_p}/Z_{\check{G}'_p}(\Lambda_p)] \leftarrow [\check{\mathcal{B}}/\check{K}]$$

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- intertwines translation functors and Jacquet restrictions/Zelevinsky derivatives

$$\begin{array}{ccc}
 \Pi(G'_p, \Lambda_p) & \sim & \Pi(G_{\mathbb{R}}, \Lambda_{\mathbb{R}}) \\
 \text{Jacquet} \downarrow & & \downarrow \text{translation} \\
 \Pi(L'_p, \Lambda'_p) & \sim & \Pi(G_{\mathbb{R}}, \Lambda'_{\mathbb{R}})
 \end{array}$$

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- intertwines translation functors and Jacquet restrictions/Zelevinsky derivatives

$$\begin{array}{ccc}
 X_{G'_p, \Lambda_p} & \sim & \check{\mathcal{B}} \\
 \uparrow \text{Lusztig ind} & & \uparrow \text{Hecke} \\
 X_{L'_p, \Lambda'_p} & \sim & \check{\mathcal{B}}
 \end{array}$$

Which pairs of groups?

The following pairs should work:

G'_p	$G_{\mathbb{R}}$
$\mathbf{GL}_m(\mathbb{Q}_p)$	$\mathbf{GL}_n(\mathbb{C})$
$\mathbf{Sp}(2m, \mathbb{Q}_p)$	$\mathbf{U}(a, b)$
$\mathbf{SO}(2m+1, \mathbb{Q}_p)$	

(each $\Lambda_{\mathbb{R}} \rightsquigarrow$ many possible (G'_p, Λ_p) 's)

The following pairs are expected to work:

$\mathbf{Sp}(2m, \mathbb{Q}_p)$	$\mathbf{Sp}(2n, \mathbb{R})$
$\mathbf{SO}(2m+1, \mathbb{Q}_p)$	$\mathbf{SO}(2n+1, \mathbb{R})$
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Automorphic representations

- G - connected reductive group $/\mathbb{Q}$
- $\mathbb{A}_{\mathbb{Q}} \subset \mathbb{R} \times \prod_p \mathbb{Q}_p$
- $G(\mathbb{A}_{\mathbb{Q}}) \hookrightarrow L^2(G(\mathbb{Q})A \backslash G(\mathbb{A}_{\mathbb{Q}}))$

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Automorphic representations: those appearing inside $L^2(G(\mathbb{Q})A \backslash G(\mathbb{A}_{\mathbb{Q}}))$.

Arthur's conjecture: classify automorphic representations.

Arthur packets

Arthur:

$$\blacksquare \{\text{automorphic reps}\} = \bigcup_{\psi} \underbrace{\Pi_{\psi}^A(G(\mathbb{A}))}_{\text{Arthur packets}}$$

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$$\blacksquare \underbrace{\pi}_{G(\mathbb{A}_{\mathbb{Q}})\text{-rep}} = \underbrace{\pi_{\mathbb{R}}}_{G(\mathbb{R})\text{-rep}} \otimes'_p \underbrace{\pi_p}_{G(\mathbb{Q}_p)\text{-rep}}$$

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$$\rightsquigarrow \Pi_{\psi}^A(G(\mathbb{A})) \approx \underbrace{\Pi_{\psi_{\mathbb{R}}}^A(G(\mathbb{R})) \text{ and } \Pi_{\psi_p}^A(G(\mathbb{Q}_p))}_{\text{local Arthur packets}}$$

Calculation of Arthur packets

Status: for G split classical,

$/\mathbb{Q}_p$ local A-packets are computable ($\exists$ algorithm)

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~~~~~ use our comparison and do calculation on the p-adic side instead (this is the reason for the desiderata...)

## Multiplicity one of A-packets

Arthur: can determine the multiplicity of  $\pi$  in  $L^2(G(\mathbb{Q})A \backslash G(\mathbb{A}_{\mathbb{Q}}))$  provided the local A-packets have “*multiplicity one*”.

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- $\psi \rightsquigarrow A(\psi)$ , finite group (usually 2-group)
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“*Multiplicity one*” means  $\dim \rho_{\pi} = 1$ .

(solved in the p-adic case for quasi-split classical groups(?))

# Local Arthur packets via geometry

Adams-Barbasch-Vogan ('92) + Adams-Arancibia-Mezo ('22):

- $\Pi_{\psi_{\mathbb{R}}}^A(G(\mathbb{R}))$ ,  $A(\psi_{\mathbb{R}})$ , and  $\rho_{\pi_{\mathbb{R}}}$  can be defined using microlocal geometry of  $[\check{B}/\check{K}]$ .

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Vogan's conjecture ('93) +

Cunningham-Fiori-Moussaoui-Mracek-Xu ('21) + ... :

- $\Pi_{\psi_p}^A(G(\mathbb{Q}_p))$ ,  $A(\psi_p)$ , and  $\rho_{\pi_p}$  can (conjecturally) be defined using microlocal geometry of  $[X_{\Lambda_p}/Z_{\check{G}}(\Lambda_p)]$ .

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Our comparison: also relates microlocal geometry info of both sides

~~~~> translate the problem to p-adic side

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Chan-Wong ('24)

$$\text{Functor: } \text{Rep}(\mathbf{GL}_n(\mathbb{C})) \xrightarrow{\Gamma_{n,m}} \text{Mod}\left(\underbrace{\mathbb{H}_m}_{\text{gr-Hecke algebra}}\right) \longrightarrow \text{Rep}(\mathbf{GL}_m(\mathbb{Q}_p)).$$

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- Construction is algebraic, and no Langlands dual
- $\approx$ our construction

Barchini-Trapa

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- Good for relating Arthur packets for the *same group*
- image is small...

Thank you!